

Introduction to Machine Learning Problems Unit

4: Model Order Selection

Shan Qureshi

September 2026

1. (a) i. Linear
ii. There is no under-modeling.
iii. True parameters 1, 2, 0
- (b) i. Non-Linear
ii. There is NO under-modeling
iii. True parameters $a_0 = 3, a_1 = 3, b_0 = 2, b_1 = 3$
- (c) i. Linear
ii. There is under-modeling

```
2.
1 import numpy as np
2 from sklearn.linear_model import LinearRegression
3
4 # \hat{y} = \sum_{j=0}^d \beta_j e^{-ju/d} where u and y
   are defined
5 N = len(u)
6 split_idx = N // 2
7
8 u_train, y_train = u[:split_idx], y[:split_idx]
9 u_test, y_test = u[split_idx:], y[split_idx:]
10
11 dtest = list(range(1, 11))
12 mse_list = []
13
14 for d in dtest:
15     X_train = np.column_stack([np.exp(-j * u_train / d)
16                                for j in range(d + 1)])
17     X_test = np.column_stack([np.exp(-j * u_test / d)
18                               for j in range(d + 1)])
19
20     model = LinearRegression()
21     model.fit(X_train, y_train)
22
23     yhat = model.predict(X_test)
24     mse = np.mean((y_test - yhat) ** 2)
```

```

23     mse_list.append(mse)
24
25 best_d = dtest[np.argmin(mse_list)]

```

3. (a) $y_i = f(x_i, \beta_0)$

$$\begin{aligned}
 \hat{\beta} &= \frac{\sum_{i=1}^N y_i}{\sum_{i=1}^N x_i^2} \\
 &= \frac{\sum_{i=1}^N \beta_0 x_i^2}{\sum_{i=1}^N x_i^2} \\
 &= \beta_0 \frac{\sum_{i=1}^N x_i^2}{\sum_{i=1}^N x_i^2} \\
 \hat{\beta} &= \beta_0
 \end{aligned}$$

Then, solving for Bias,

$$\begin{aligned}
 Bias(x) &:= \mathbb{E}[f(x, \hat{\beta})] - f(x, \beta_0) \\
 &= \mathbb{E}[\beta_0 x^2] - \beta_0 x^2 \\
 &= 0
 \end{aligned}$$

(b) $y_i = f(x_i, \beta_0) + \epsilon_i$ and $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$

$$\begin{aligned}
 \hat{\beta} &= \frac{\sum_{i=1}^N (\beta_0 x_i^2 + \epsilon_i)}{\sum_{i=1}^N x_i^2} \\
 &= \beta_0 + \frac{\sum_{i=1}^N \epsilon_i}{\sum_{i=1}^N x_i^2}
 \end{aligned}$$

Then, solving for Bias,

$$\begin{aligned}
 Bias(x) &= \mathbb{E} \left[\left(\beta_0 + \frac{\sum_{i=1}^N \epsilon_i}{\sum_{i=1}^N x_i^2} \right) x^2 \right] - \beta_0 x^2 \\
 &= \mathbb{E}[\beta_0 x^2] + \mathbb{E} \left[\frac{\sum_{i=1}^N \epsilon_i}{\sum_{i=1}^N x_i^2} \right] x^2 - \beta_0 x^2 \\
 &= 0
 \end{aligned}$$

(c) $y_i = f(x_i + \epsilon_i, \beta_0)$

Expanding y_i ,

$$\begin{aligned}
 y_i &= \beta_0(x_i + \epsilon_i)^2 \\
 &= \beta_0(x_i^2 + 2x_i\epsilon_i + \epsilon_i^2) \\
 \mathbb{E}[y_i] &= \beta_0(x_i^2 + 0 + \sigma^2) \\
 \mathbb{E}[\hat{\beta}] &= \frac{\sum_{i=1}^N \mathbb{E}[y_i]}{\sum_{i=1}^N x_i^2} \\
 &= \frac{\sum_{i=1}^N \beta_0(x_i^2 + \sigma^2)}{\sum_{i=1}^N x_i^2} \\
 &= \beta_0 + \beta_0 \frac{N\sigma^2}{\sum_{i=1}^N x_i^2}
 \end{aligned}$$

Then our Bias is calculated as,

$$\begin{aligned}
 Bias(x) &= (\mathbb{E}[\hat{\beta}] - \beta_0)x^2 \\
 &= \left(\beta_0 + \beta_0 \frac{N\sigma^2}{\sum_{i=1}^N x_i^2} - \beta_0 \right) x^2 \\
 &= \left(\beta_0 \frac{N\sigma^2}{\sum_{i=1}^N x_i^2} \right) x^2
 \end{aligned}$$

4. (a) $\hat{\beta} = (X^T X)^{-1} X^T y$
 (b) We let $y = X_0 \beta_0$, then $\hat{\beta} = (X^T X)^{-1} X^T X_0 \beta_0$.
 (c)

```

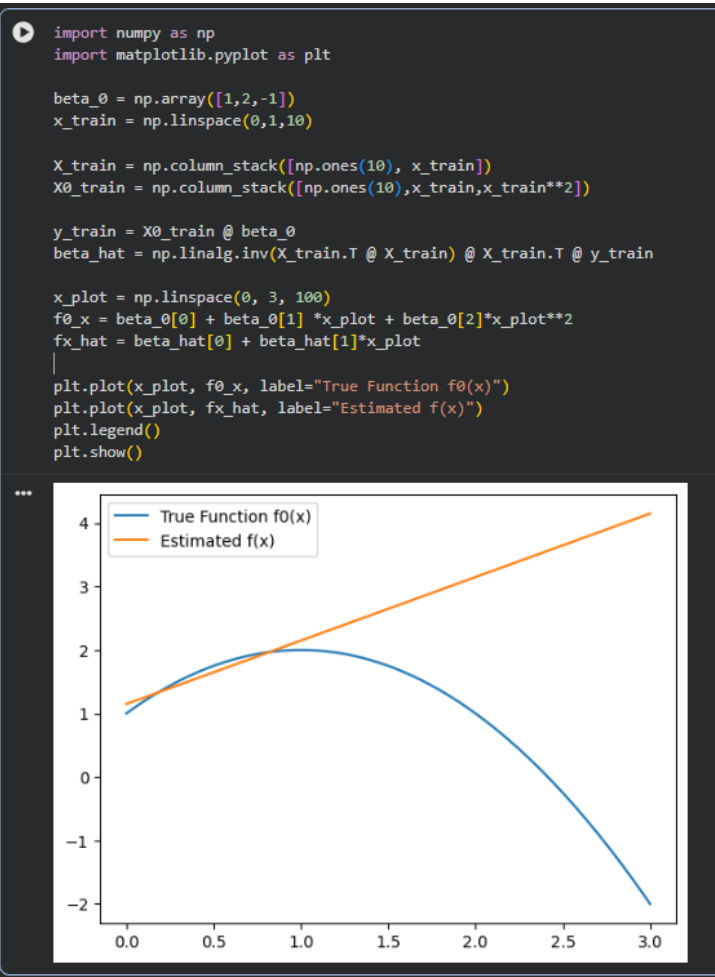
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 beta_0 = np.array([1,2,-1])
5 x_train = np.linspace(0,1,10)
6
7 X_train = np.column_stack([np.ones(10), x_train])
8 X0_train = np.column_stack([np.ones(10), x_train,
9                               x_train**2])
10
11 y_train = X0_train @ beta_0
12 beta_hat = np.linalg.inv(X_train.T @ X_train) @
13             X_train.T @ y_train
14
15 x_plot = np.linspace(0, 3, 100)
16 f0_x = beta_0[0] + beta_0[1] * x_plot + beta_0[2] *
17         x_plot**2
18 fx_hat = beta_hat[0] + beta_hat[1] * x_plot
19 plt.plot(x_plot, f0_x, label="True Function f0(x)")

```

```

18 plt.plot(x_plot, fx_hat, label="Estimated f(x)")
19 plt.legend()
20 plt.show()

```



- (d) The bias is largest at $x = 3$
5. (a) **Model 1:** $\hat{y} = \beta_0 + \beta_1 v$
Model 2: $\hat{y} = \beta_0 + \beta_1 v + \beta_2 a$
Model 3: $\hat{y} = \beta_0 + \beta_1 (v \cdot t_1) + \beta_2 (v \cdot t_2) + \beta_3 a$
- (b) Model 1 has 2 parameters.
Model 2 has 3 parameters.
Model 3 has 4 parameters and is the most complex.

(c) i. **Model 1:**

$$A = \begin{bmatrix} 1 & 0.7 \\ 1 & 1.3 \\ 1 & 1.6 \end{bmatrix}$$

ii. **Model 2:**

$$A = \begin{bmatrix} 1 & 0.7 & 55 \\ 1 & 1.3 & 65 \\ 1 & 1.6 & 70 \end{bmatrix}$$

iii. **Model 3:**

$$A = \begin{bmatrix} 1 & 0.7 & 0 & 55 \\ 1 & 1.3 & 0 & 65 \\ 1 & 1.6 & 0 & 70 \end{bmatrix}$$

(d) The standard error for model 3 is $\frac{0.05}{\sqrt{10}} \approx 0.0158$. Model 2 has a Mean Test RSS of 0.72 which is greater than Model 3 one standard rule bound of $0.7 + 0.0158 = 0.7158$.